NIKHIL CHAGANTI

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PROFESSIONAL EXPERIENCE

IGATE SOLUTIONS – Software Engineer – Frisco, Texas, United States
(January, 2025 – Present)

- **Developed and maintained a dynamic, responsive front-end user interface** using ReactJS, HTML, CSS, and JQuery, improving customer interaction and increasing engagement by 30%.
- **Designed and integrated RESTful APIs using JSON** for seamless data communication between UI and backend services, enabling real-time updates and enhanced user experiences.
- **Optimized client-side performance** using React hooks and component reusability, reducing page load times by 40% and improving application responsiveness.
- **Created interactive dashboards and visualizations** using Power BI, Tableau, Seaborn, and Pandas to support real-time decision-making and internal analytics.
- **Integrated cloud-based data pipelines** with AWS Redshift, S3, and AWS Glue for scalable, secure, and efficient data access and transformation workflows.
- **Implemented robust ETL pipelines** using PySpark, SparkSQL, Apache Spark, MapReduce, and Hadoop to process and transform large datasets for front-end display and analytics.
- **Leveraged Databricks and SAS Frameworks** for collaborative data science workflows and enterprise-grade analytics integration.
- **Used Git for version control and CI/CD pipelines**, ensuring consistent deployment, team collaboration, and high code quality.

RESEARCH ANALYST – UNIVERSITY OF MARYLAND, BALTIMORE COUNTY
(January, 2023 – December, 2024)

- **Designed and implemented a sentiment-based tweet recommendation system** analyzing 1.6 million tweets using NLP techniques such as lemmatization, stop-word removal, and text normalization, achieving an **85% classification accuracy**.
- **Conducted extensive data preprocessing**, including null value imputation and duplicate removal, to ensure data integrity for sentiment analysis and model training.
- **Built dynamic visual dashboards** using Matplotlib, Seaborn, and Plotly to showcase sentiment distribution, temporal trends, and keyword frequency, supporting real-time data interpretation and insights for social media engagement strategies.
- **Developed a data mining model to predict high-risk road accident zones** using a historical dataset of 145,000+ records, attaining **90% model accuracy** through spatial analysis and refined feature engineering.
- **Identified key accident patterns** by applying outlier detection and temporal-spatial clustering, leading to actionable safety insights such as optimal placement of speed control measures.
- **Built interactive dashboards** with Plotly and Dash to visualize accident hotspots, severity levels, and time-based incident trends, enabling stakeholders to make informed decisions for public safety initiatives.
- **Applied machine learning models and data processing** using Scikit-Learn, TensorFlow, and PyTorch for predictive insights and analytics-ready datasets.

COGNIZANT TECHNOLOGY SOLUTIONS – Programmer Analyst – Hyderabad, India
JP Morgan Chase (February, 2022 – October, 2022)

- Gained hands-on experience working on the JP Morgan Chase investment banking project, contributing to **data security** initiatives.
- Completed **specialized training in data security**, enhancing understanding of **risk management** in financial services.
- Acquired foundational skills in **SQL, ETL processes, and AI & Analytics (AIA) domains**, applied in real-world projects.

- Utilized **Informatica Power Center** to design and manage ETL workflows, ensuring efficient data extraction, transformation, and loading from multiple sources into the **Data Warehouse**, supporting secure and scalable reporting systems.
- Strengthened client management and vendor relations skills, contributing to successful project outcomes.
- Developed expertise in **phishing mail detection** and cybercrime analysis as part of the **CDB AI & Analytics** service line at Cognizant.
- Earned **EF SET certification**, demonstrating proficiency in English language communication and analysis.
- Completed a 9-month tenure with recognition and a service letter from Cognizant for outstanding performance.

TECHNICAL SKILLS

Programming Languages & Tools	Python, C, R
Database Management	MySQL, ETL pipelines, Data Warehouse, PL/SQL
Data Analysis	Data Mining, Machine Learning, Power BI, Tableau, Weka, NLP Technologies, Informatica Power Center
AWS Cloud Technologies	S3, Redshift, Lambda, Glue, EC2, API, Snowflake, IAM
Web Technologies	HTML, CSS, React JS, JSON, JQUERY, UI
Tools	Jupyter Notebook, Visual studio Code, GitHub
Frameworks	PySpark, SparkSql, Databricks, Hadoop, Sckit-Learn, Apache Spark, Pandas, Tensor Flow, Numpy, Pytorch, Map Reduce, SAS Framework, Seaborn
Methodologies & Cloud	SDLC, CRM, ERP Systems, GCP

PROJECTS:

- **Sentiment Analysis of E-Commerce Company Product Reviews**
  Implemented a Decision Support System (DSS) to analyze 189,874 product reviews from Company, achieving 88% accuracy in sentiment classification. Utilized TextBlob for sentiment scoring and trained models including Logistic Regression, LinearSVC, Random Forest, and Multinomial Naive Bayes. Logistic Regression provided the highest accuracy in classifying customer feedback. Generated a word cloud using Weka to visualize the most frequent and impactful terms in customer reviews, enhancing insights into key sentiment-driving words.
- **Customer behaviour Analysis with AWS Data pipeline**
  Developed a data pipeline using AWS services to analyze customer behaviour for an e-commerce platform. The pipeline involved extracting raw transaction data from Amazon S3, processing it with AWS Glue for data cleaning and transformation, and storing the results in Amazon Redshift for querying. Used SQL to analyze customer purchasing patterns and generate insights, which were visualized using Tableau. This project improved data processing efficiency, reduced report generation time by 30%, and provided actionable insights into customer preferences and sales trends, enhancing business decision-making.
- **Todo List using React js**
  Developed a dynamic To-Do List application using React.js, utilizing useState for state management to handle task creation, updates, and deletion. Implemented component-based architecture with props, enabling reusable and maintainable UI components such as TodoItem, TodoForm, and TodoList. Enhanced UI with FontAwesome icons, integrating interactive elements like task deletion (faTrash) and completion toggles (faPenToSquare) for an intuitive user experience. Utilized UUID for unique task identifiers, ensuring each task has a distinct key to avoid rendering issues and improve application performance. Leveraged React hooks (useState) to manage task states dynamically, enabling smooth state transitions without reloading the page.

CERTIFICATIONS:

- AWS Certified Data Engineer – Associate
- Oracle Database SQL certified - Associate
- Cisco Certified Python – Programming Essentials in Python
- Certified in Artificial Intelligence Fundamentals by IBM Watson

EDUCATION:

- **Master of Science in Information Systems -** CGPA – 3.93/4.0
  University of Maryland, Baltimore County, Baltimore, MD, USA (January, 2023 – December, 2024)
- **Bachelor of Technology in Electronics and Communication Engineering -** CGPA – 8.23/10.0
  SRKR Engineering College, Andhra Pradesh, India (2018 – 2022)